

WITTENBERG, Germany

WMO: 104740

Lat: 51.8892N

Lon: 12.6444E

Elev: 347

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	13.6	2.0	6.2	11.0	7.4	8.2	14.7	26.0	43.9	23.4	42.1	5.2	50	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	86.0	66.0	82.4	64.9	79.3	63.2	68.1	81.6	66.3	78.0	64.8	75.8	7.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	89.7	70.9	62.1	84.3	70.2	60.4	79.6	69.8	32.6	81.6	31.1	77.9	30.0	76.3	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.5	16.1	DB	6.7	93.5	8.8	3.5	0.3	96.0	-4.9	98.0	-9.8	100.0	-16.3	102.5	(4)
(5)			WB	5.8	72.8	8.7	2.6	-0.5	74.7	-5.6	76.3	-10.4	77.7	-16.7	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	32.7	32.0	40.1	46.5	56.2	61.2	65.6	65.0	57.8	49.6	40.0	35.1
	DBStd	13.96	10.13	9.35	7.09	6.64	6.94	6.56	6.19	6.05	5.37	6.87	6.64	7.30
	HDD50	2383	538	505	312	145	24	0	0	0	3	91	304	461
	HDD65	6218	1003	924	773	555	283	153	70	78	225	478	751	925
	CDD50	1867	1	1	5	40	215	336	484	465	237	79	3	1
	CDD65	226	0	0	0	0	9	39	89	78	10	1	0	0
	CDH74	2183	0	0	0	2	109	397	893	684	93	5	0	0
	CDH80	589	0	0	0	0	3	102	274	199	11	0	0	0
Wind	WSAvg	7.0	8.6	7.7	7.9	7.2	6.4	6.1	6.0	5.5	6.2	6.9	7.5	7.8
Precipitation	PrecAvg	24.2	2.0	1.5	1.7	1.3	2.2	2.3	2.9	2.4	2.0	1.6	2.0	2.0
	PrecMax	32.7	3.7	3.1	4.4	3.6	4.9	5.4	7.2	5.5	5.7	3.5	5.2	4.0
	PrecMin	16.8	0.2	0.4	0.3	0.0	0.4	0.7	0.4	0.9	0.4	0.3	0.0	0.7
	PrecStd	4.3	0.9	0.7	1.1	0.8	1.3	1.1	1.5	1.3	1.3	0.9	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	58.0	63.5	72.5	80.3	88.6	89.9	91.6	81.7	72.9	57.9	54.0
		MCWB	48.6	50.9	52.6	56.2	61.5	66.7	67.9	69.0	64.3	61.3	52.2	49.3
	2%	DB	48.8	51.9	57.6	66.5	77.2	82.6	86.4	85.0	76.4	67.0	54.2	51.0
		MCWB	45.5	47.0	49.9	53.6	60.0	64.8	66.0	66.3	62.2	57.7	50.5	47.3
	5%	DB	46.6	46.8	53.3	62.7	73.9	78.4	83.0	81.2	72.1	62.9	51.5	48.5
		MCWB	43.6	43.1	47.1	51.7	58.5	62.0	65.3	64.7	60.9	55.9	48.0	45.9
	10%	DB	44.3	43.3	50.2	58.7	69.7	74.6	79.2	77.5	67.9	59.8	49.1	45.5
		MCWB	41.3	40.0	45.1	49.5	56.9	61.1	63.6	63.3	58.7	53.7	46.0	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	51.4	53.8	57.2	64.8	68.4	71.9	71.1	65.8	62.2	53.5	49.5
		MCDB	52.1	57.3	60.7	68.9	76.3	83.5	86.4	87.9	77.4	70.4	56.3	52.7
	2%	WB	46.2	47.1	50.5	54.6	62.2	65.8	68.5	67.8	63.8	58.9	50.9	47.6
		MCDB	48.4	51.1	56.7	65.3	72.9	77.9	81.8	80.8	73.0	65.4	53.9	50.3
	5%	WB	43.9	43.6	47.9	52.3	59.9	64.1	66.4	66.2	62.0	56.4	48.5	45.8
		MCDB	46.3	46.7	52.0	60.8	70.1	75.2	78.2	77.5	69.3	62.2	51.2	48.5
	10%	WB	41.5	40.3	45.5	50.1	57.8	61.9	64.7	64.6	60.1	54.2	46.2	43.1
		MCDB	44.3	43.1	49.7	57.7	68.0	72.4	76.2	74.6	67.0	58.7	48.8	45.3
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.9	13.1	16.3	18.3	17.3	18.8	18.7	16.0	14.1	9.7	7.9
		MCDBR	9.1	14.1	18.9	22.7	24.5	24.5	25.5	25.0	22.4	19.1	12.5	10.1
		MCWBR	8.1	10.5	12.6	12.4	11.7	11.0	11.0	11.0	11.2	11.6	9.9	8.2
	5% WB	MCDBR	9.6	14.1	17.4	20.9	22.2	21.7	22.7	21.7	20.2	17.9	12.3	9.4
		MCWBR	8.9	10.8	12.2	11.8	11.6	10.5	11.4	10.3	11.2	11.5	10.1	8.2
Clear-Sky Solar Irradiance	taub	0.318	0.341	0.382	0.415	0.416	0.412	0.432	0.425	0.387	0.365	0.341	0.312	
	taud	2.354	2.333	2.235	2.200	2.240	2.277	2.231	2.266	2.350	2.385	2.390	2.362	
	Ebn at Noon	216	244	254	259	266	267	259	254	251	232	205	197	
	Edh at Noon	20	26	35	41	41	40	42	38	31	25	19	17	
All-Sky Solar Radiation	RadAvg	219	437	758	1256	1546	1648	1551	1340	970	553	260	166	
	RadStd	26	75	91	161	185	133	200	113	117	65	36	18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A

Nomenclature: See separate page